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Degree: B.Tech. Semester: III - Course work

MID-SEMESTER EXAMINATION, SEPTEMBER 2026

Course Title: *Mathematics for Signal Processing and Communication*

Course Code: ECMTC04

Time: 1:30 Hours **Max. Marks: 25**

Note: Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1*a*	Prove that the set $H = \{A \in M_3(\mathbb{R}) : \text{trace}(A) = 0\}$ forms a subgroup of the group $(M_3(\mathbb{R}), +)$ of all 3×3 real matrices under matrix addition.	2.5	CO1
1*b*	Let $$G = \left\{ \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} : a \in \mathbb{Z}_6 \right\}$$ be a group under matrix multiplication. Determine the order of the group G. Also, find the order of each element of G.	2.5	CO1
2*a*	Define a cyclic group. Determine whether the group $U(11)$ is cyclic. Justify your answer.	2.5	CO1
2*b*	Let $f : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$ be defined by $f(x) = 3x \pmod{12}$. Prove that f is a group homomorphism. Hence, find $\ker(f)$.	2.5	CO1
3*a*	Define the term **Integral Domain**. Show that Z_5 forms an integral domain. Whether Z_5 also forms a field?	2.5	CO1
3*b*	Determine whether the set $W = \{(x, y, z) \in \mathbb{R}^3 \mid 13x + 5y = 2z\}$ forms a subspace of $\mathbb{R}^3$.	2.5	CO2
4a	Investigate the linear independence or dependence of the set of vectors $x_1 = (2, 0, 0)$, $x_2 = (2, 2, 2)$, $x_3 = (0, 0, 1)$, $x_4 = (0, 1, 0)$ in $\mathbb{R}^3$.	2.5	CO2
4*b*	Consider the vector space $\mathbb{P}_2$ of polynomials of degree at most 2. Is the set $\{1 + x, x + x^2, 1 + x^2\}$ forms a spanning set for the vector space $\mathbb{P}_2$? Justify your answer.	2.5	CO2
5*a*	Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(a_1, a_2) = (2a_1 + 3a_2, 2a_1 - a_2)$. Show that T is a linear transformation.	2.5	CO2
5*b*	Using Rank Nullity Theorem, show that the linear transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ given by $T(x, y, z) = (x - y, y - z, z - x, 0)$ is not bijective.	2.5	CO2